

3) Week 12: Bisection and Newton-Raphson

3.1) Under what conditions might the bisection method fail to find a root of $f(x)$? What about Newton-Raphson?

- Bisection may fail if:
 - > $f(x)$ has no roots.
 - > $f(x)$ is not continuous
- Newton Raphson may fail if
 - > $f(x)$ has no roots
 - > $f(x)$ is not differentiable
 - > $f(x)$ has locations where $f'(x)=0$

3.2) By what factor does the interval that contains the root decrease in each iteration of the bisection method?

By a factor of 2

3.3) Carry out the first three iterations of the Newton-Raphson method for the function $f(x) = x^2 - 4$ with initial value $x_0 = 6$.

$$f'(x) = 2x$$

$$\text{Newton Raphson: } x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

$$= x_k - \frac{x_k^2 - 4}{2x_k}$$

$$= \frac{x_k}{2} + \frac{2}{x_k} \quad \frac{68}{30} = \frac{34}{15}$$

$$x_1 = \frac{x_0}{2} + \frac{2}{x_0} = 3 + \frac{1}{3} = \frac{10}{3}$$

$$x_2 = \frac{x_1}{2} + \frac{2}{x_1} = \frac{10}{6} + \frac{6}{10} = \frac{100 + 36}{60} = \frac{136}{60} = \frac{34}{15}$$

$$x_3 = \frac{136}{120} + \frac{120}{136} = 2.02.$$

3.4) How many iterations will be taken by the bisection method to find a root of the function $f(x) = x^2 - 11$ if the initial interval is $[0, 5]$ and the tolerance is 0.5?

$$K_{\min} = \text{ceil } \log_2 \frac{5-0}{0.5} = 4$$

3.5) You wish to use the bisection method to find a root of $f(x) = x^2 - 3x + 1$ to within a tolerance of 10^{-3} .

- Suggest a viable initial condition for the iteration.
- Compute the interval after two iterations.
- How many iterations will it take for the algorithm to converge given your suggested initial condition?

a) Need N_0 and P_0 such that $f(N_0) < 0$ and $f(P_0) > 0$.

$$P_0 = 0 \rightarrow f(P_0) = 1 \quad \checkmark$$

$$N_0 = 1 \rightarrow f(N_0) = 1 - 3 + 1 = -1 \quad \checkmark$$

b)

k	N_k	P_k	C	$f(C)$
0	1	0	0.5	$0.25 - 1.5 + 1 = -0.25 < 0$
1	0.5	0	0.25	$0.25^2 - 0.75 + 1 = 0.3125 > 0$
2	0.5	0.25	0.375	

c)

$$k_{\min} = \text{ceil} \log_2 \frac{1-0}{10^{-3}} = 10$$

3.6) You wish to use the bisection method to find a root of $f(x) = e^{-x} \cos(x)$ in the interval $[0, 3]$.

- Is $[0, 3]$ a viable initial interval for the iteration? If not, suggest another.
- Create a table with the first three iterations of the method.
- How many iterations are needed to find a root to within a tolerance of 10^{-5} ?

$$\begin{aligned} a) \quad f(0) &= e^{-0} \cos 0 = 1 > 0 \\ f(3) &= e^{-3} \cos 3 = -0.05 < 0 \end{aligned}$$

We have a negative and a positive evaluation, so yes.

b)

k	N_k	P_k	C	$f(C)$
0	3	0	1.5	0.1578 > 0
1	3	1.5	2.25	-0.0662 < 0
2	2.25	1.5	1.875	-0.046 < 0
3	1.875	1.5	1.6875	-0.021

$$c) \quad k_{\min} = \text{ceil} \log_2 \frac{3}{10^{-5}} = 19$$

3.7) You wish to find a root of the function $f(x) = e^{-x} - x$, defined on the real line.

- Will Newton-Raphson converge to the root regardless of the initial condition? Why or why not?
- Carry out the first two iterations of Newton-Raphson, starting from $x_0 = 1$.

a) We need $f(x)$ to be continuous and never have $f'(x) = 0$.

- $f(x)$ is continuous everywhere ✓

- $f'(x) = -e^{-x} - 1 = 0 \iff e^{-x} = -1 \dots$ impossible.

$\therefore$ We expect N.R. to converge to a root, if one exists, for any x_0 .

$$\begin{aligned} \text{b) } x_{k+1} &= x_k - \frac{f(x_k)}{f'(x_k)} = x_k + \frac{e^{-x_k} - x_k}{e^{-x_k} + 1} \\ \therefore &= \frac{x_k e^{-x_k} + \cancel{x_k} + e^{-x_k} - \cancel{x_k}}{e^{-x_k} + 1} \\ &= \frac{x_k + 1}{e^{x_k} + 1} \end{aligned}$$

$$x_1 = \frac{2}{e+1} = 0.5379$$

$$x_2 = \frac{0.5379 + 1}{e^{0.5379} + 1} = 0.5669$$

3.8) You wish to find a root of the function $f(x) = x^3 - 3x + 1$ using the Newton-Raphson method. Give an initial condition for which the approach is guaranteed to fail.

It will fail if $f'(x_0) = 0$.

$$f'(x) = 3x^2 - 3$$

$$f'(x) = 0 \Leftrightarrow 3x^2 - 3 = 0$$

$$\Leftrightarrow x^2 - 1 = 0$$

$$\Leftrightarrow x = \pm 1$$

3.9) You wish to use the bisection method to find a root of the function:

$$f(x) = (x - 2.5)(x + 1)(x - 1)$$

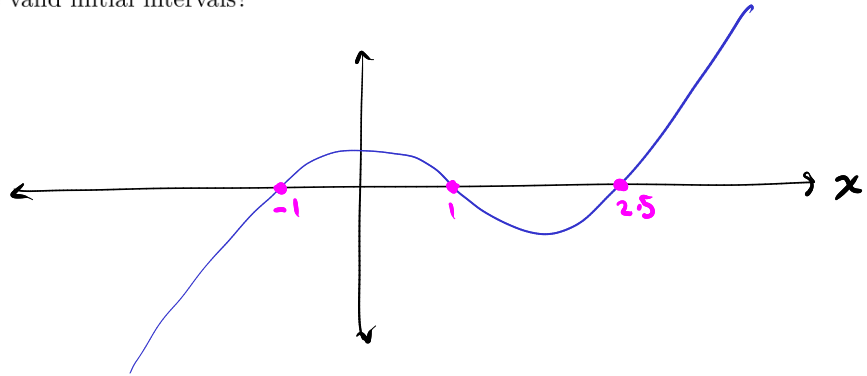
Which of the following options are valid initial intervals?

☐ $[-1, 1]$

☐ $[0, 1]$

☐ $[0, 2]$

☐ $[0, 3]$



For an initial interval $[a, b]$ to be valid we need
either $f(a) > 0$, $f(b) < 0$ or $f(b) < 0$, $f(a) > 0$.

• $[-1, 1]$: $f(a) = f(b) = 0$ X

• $[0, 1]$: $f(0) > 0$, $f(1) = 0$ X

• $[0, 2]$: $f(0) > 0$, $f(2) < 0$ ✓

• $[0, 3]$: $f(0) > 0$, $f(3) > 0$ X

4) Week 12: Fixed points

4.1) We wish to find a root of the function

$$f(x) = x^2 - \cos x$$

a) Which of the following functions g have a fixed point that is a root of f ?

☐ $g_1(x) = x + x^2 - \cos x$

☐ $g_2(x) = \sqrt{\cos x}$

☐ $g_3(x) = x - \cos x$

• x is fixed point of $g_1 \Rightarrow \cancel{x + x^2 - \cos x} = \cancel{x}$

$\Rightarrow x^2 - \cos x = 0 \Rightarrow f(x) = 0$

✓

• x is fixed point of $g_2 \Rightarrow \sqrt{\cos x} = x$

$\Rightarrow \cos x = x^2 \Rightarrow f(x) = 0$

✓

• x is fixed point of $g_3 \Rightarrow x - \cos x = x$

$\Rightarrow \cos x = 0 \not\Rightarrow f(x) = 0$

✗

For example $x = \frac{\pi}{2}$ is f.p. of g_3 but not root of f .

b) Suppose that we know that a root exists in the interval $[0.5, 1]$. For which functions g is the fixed point iteration guaranteed to converge when initialized in that interval? (the plot of $\frac{\sin x}{\sqrt{\cos x}}$ is shown for reference).

☐ $g_1(x) = x + x^2 - \cos x$

☐ $g_2(x) = \sqrt{\cos x}$

☐ $g_3(x) = x - \cos x$

• $g_1: \quad g'_1(x) = 1 + 2x + \sin x$

$$\max_{[0.5, 1]} |g'_1(x)| = \max_{[0.5, 1]} \underbrace{1 + 2x}_{\text{positive}} + \underbrace{\sin x}_{\text{positive}} > 1$$

✗

This is not guaranteed to converge.

• $g_2: \quad g'_2(x) = -\frac{1}{2} \frac{\sin x}{\sqrt{\cos x}}$

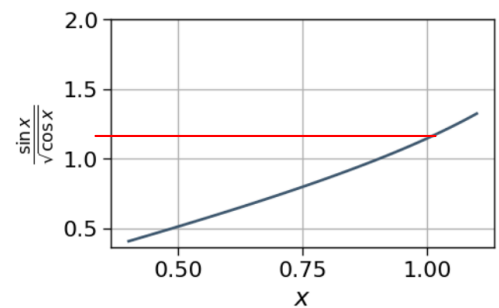
For $\max_{[0.5, 1]} |g'_2(x)| < 1$ to

hold, we need

$$\left| -\frac{1}{2} \frac{\sin x}{\sqrt{\cos x}} \right| < 1 \quad \forall x \in [0.5, 1]$$

Or, what is the same,

$$\left| \frac{\sin x}{\sqrt{\cos x}} \right| < 2 \quad \forall x \in [0.5, 1]$$



The figure shows that this is true.

$\therefore$ FPI will converge for any initial point inside $[0.5, 1]$

✓

4.2) You wish to find a root of the function

$$f(x) = x^3 - 2x + 1$$

using FPI with the auxiliary function

$$g(x) = \frac{x^3 + 1}{2}$$

- a) Prove that all of the fixed points of $g(x)$ are roots of $f(x)$.
- b) Supposing we know that a fixed point of g exists in the interval $[0, 0.7]$, prove that FPI will converge for any initial condition in that interval.
- c) Create a table with the first two iterations of the fixed point iteration starting from $x_0 = 0$

a) x is a f.p. of $g(x) \Rightarrow \frac{x^3+1}{2} = x \Rightarrow x^3 - 2x + 1 = 0$
 $\Rightarrow x$ is root of $f(x)$ ✓

b) $g'(x) = \frac{3}{2} x^2$

$\therefore \max_{[0, 0.7]} \frac{3}{2} x^2 = \frac{3}{2} (0.7)^2 = 0.735 < 1$ □

c)

k	x_k	$g(x_k)$
0	0	$\frac{1}{2}(0^3+1) = \frac{1}{2}$
1	$\frac{1}{2}$	$\frac{1}{2}\left(\left(\frac{1}{2}\right)^3+1\right) = \frac{9}{16}$
2	$\frac{9}{16}$	$\frac{1}{2}\left(\left(\frac{9}{16}\right)^3+1\right) = 0.589$

4.3) You wish to use a fixed-point iteration to find a root of the function $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$

$$f(x_1, x_2) = \left(-x_1 + \frac{x_2}{2} + 1, x_1 x_2 - 1 \right)$$

You consider the auxiliary function $g: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ (with $x_1 \neq 0$)

$$g(x_1, x_2) = \left(\frac{x_2}{2} + 1, \frac{1}{x_1} \right)$$

- a) Is this a valid choice, in the sense that the fixed points of g are roots of f ?
- b) Find a region of $\mathbb{R}^2$ where the fixed point iteration is guaranteed to converge to a solution, provided one exists in that region.
- c) Pick an initial condition in the region you just found and carry out the first two iterations of FPI.

a) (x_1, x_2) is a f.p. of $g \Rightarrow g(x_1, x_2) = (x_1, x_2)$

$$\Rightarrow \left(\frac{x_2}{2} + 1, \frac{1}{x_1} \right) = (x_1, x_2)$$
$$\Rightarrow \left(\frac{x_2}{2} + 1 - x_1, \frac{1}{x_1} - x_2 \right) = (0, 0)$$
$$\Rightarrow \left(\frac{x_2}{2} + 1 - x_1, 1 - x_1 x_2 \right) = (0, 0)$$
$$\Rightarrow \left(\frac{x_2}{2} + 1 - x_1, x_1 x_2 - 1 \right) = (0, 0)$$
$$\Rightarrow f(x_1, x_2) = (0, 0) \quad \checkmark$$

b) We want to find a region of $\mathbb{R}^2$ where all partial derivatives have absolute value less than 1.

$$\bullet \frac{\partial g_1}{\partial x_1} = \frac{\partial}{\partial x_1} \left(\frac{x_2}{2} + 1 \right) = 0 \quad \dots \text{ok everywhere}$$

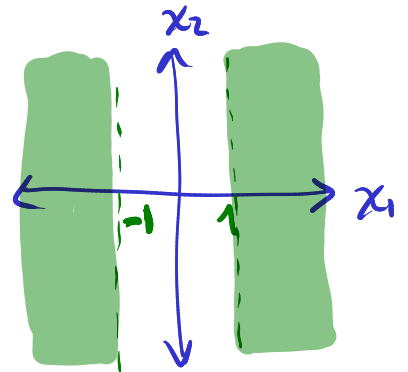
$$\bullet \frac{\partial g_1}{\partial x_2} = \frac{\partial}{\partial x_2} \left(\frac{x_2}{2} + 1 \right) = \frac{1}{2} \quad \dots \text{ok everywhere.}$$

- $\frac{\partial g_2}{\partial x_1} = \frac{\partial}{\partial x_1} \left(\frac{1}{x_1} \right) = -\frac{1}{x_1^2}$

$$\left| \frac{\partial g_1}{\partial x_1} \right| < 1 \iff \frac{1}{x_1^2} < 1 \iff x_1^2 > 1 \iff \begin{matrix} x_1 > 1 \\ \text{or} \\ x_1 < -1 \end{matrix}$$

- $\frac{\partial g_2}{\partial x_2} = \frac{\partial}{\partial x_2} \left(\frac{1}{x_1} \right) = 0 \dots \text{ok everywhere.}$

$\therefore$ FPI will converge to a root of f whenever initialized with $x_1 > 1$ or $x_1 < -1$ provided a root of f exists in that region



4.4) Consider the function $g(x) = 4x^3 + x^2$.

a) What are the fixed points of $g(x)$?

b) Create a table with the first three iterations of FPI with initial value $x_0 = -0.5$.

~~c) Find an interval for which it is guaranteed that if you start FPI in that interval, it will converge to a positive fixed point of $g(x)$~~ **REMOVED**

a) Fixed points of g have $g(x) = x$

$$4x^3 + x^2 = x$$

$$\Leftrightarrow x(4x^2 + x - 1) = 0.$$

$$\Leftrightarrow \begin{cases} x=0 \\ \text{or} \\ 4x^2 + x - 1 = 0 \end{cases}$$

$$\Leftrightarrow \begin{cases} x=0 \\ \text{or} \\ x = \frac{-1 \pm \sqrt{17}}{8} \end{cases}$$

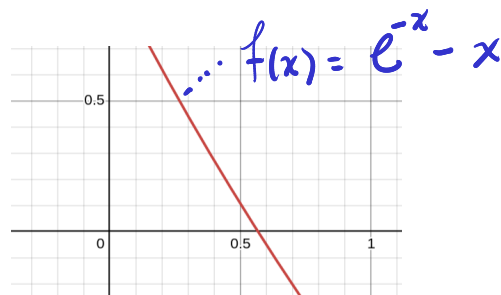
b)

k	x_k	$g(x_k)$
0	-0.5	$4 \cdot (-0.5)^3 + 0.5^2 = -0.25$
1	-0.25	$4 \cdot (-0.25)^3 + 0.25^2 = 0$
2	0	$4(0)^3 + 0^2 = 0$

4.5) We wish to find a root of the function $f(x) = e^{-x} - x$, defined on the real line. Suppose that we know that a root of f exists in the interval $[0.5, 1]$. Prove that $g(x) = e^{-x}$ is a good proxy function for finding this root using a fixed point iteration in the sense that a) a fixed point of $g(x)$ is a root of $f(x)$ and b) the fixed point iteration is guaranteed to converge.

$$f(x) = e^{-x} - x$$

$$g(x) = e^{-x}$$



a)

$$x \text{ is a fixed point of } g \Rightarrow g(x) = x$$

$$\Rightarrow e^{-x} = x$$

$$\Rightarrow e^{-x} - x = 0 \Rightarrow f(x) = 0$$

$$\Rightarrow x \text{ is root of } f.$$

b) We want to prove $\max_{x \in [0.5, 1]} |g'(x)| < 1$

$$\text{With } g(x) = e^{-x} \text{ we have } |g'(x)| = e^{-x}$$

$$\therefore \max_{x \in [0.5, 1]} |g'(x)| = \max_{x \in [0.5, 1]} e^{-x}$$

$$= e^{-0.5} = 0.6 < 1$$

